J.A.R.V.I.S.

Setup guide: Claude-powered assistant with phone chat, phone calls, and a desktop HUD

Do this first

Revoke the API key you pasted into the chat earlier. Anything pasted into a chat is stored and should be treated as public. Go to **platform.claude.com > API Keys**, delete it, and create a new one.

From now on, keys only ever go into Jarvis's settings tool (Step 3). Never into chats, emails, or GitHub.

How it fits together

Jarvis is a small program that runs all the time on a computer you control. It sends what you say to Claude, runs tools (web search, texting you, calling you), and answers. You reach it three ways:

Channel	What it is	Needs
Telegram	Text Jarvis from your phone, any time	Free Telegram bot
Phone call	Jarvis rings your cell and you talk to it	Twilio number + a tunnel (ngrok)
Desktop HUD	The movie-style screen with voice, at your PC	A Windows PC with Chrome or Edge

Pick your path

	Path A: Cloud server	Path B: Home PC
When	Now, set up from a laptop	When you have a PC to leave on
You get	Telegram + phone calls, always on	Everything, including the HUD with voice and PC control
Hosting cost	About $0-4 per month	$0 (it uses your PC)
Hardest part	Clicking through Google's cloud console	Nothing; double-click an installer

Recommendation

Start with **Path A, Telegram only**. It's the cheapest way to have a working Jarvis today. Add phone calls once you're happy, and move to Path B when you have a PC. Your settings carry over.

What it costs (approximate, check current prices)

Item	Price	Notes
Claude API	Pay per use	Sonnet 5.5: $2 / $10 per million input / output tokens. Opus 5.5: $4 / $20. A short back-and-forth on Sonnet is typically well under a cent. Prepaid credit; you set a monthly limit.
Telegram	Free	
Cloud server	$0-4 / month	Google Cloud's e2-micro VM is free, but Google bills its public IP (about $0.005/hour, roughly $3.65/month). Oracle's Always Free tier is $0, but it stops servers that sit idle for a week, which Jarvis usually does.
Twilio (calls)	$1.15 / month + per minute	Local US number $1.15/month; outbound US calls about $0.013-0.014/minute, plus speech-recognition fees. New accounts get trial credit.
ngrok (tunnel)	Free	Free plan includes one fixed web address, which is all Jarvis needs.

Realistic budget for light personal use: about $5-10 per month on Sonnet with occasional calls. This is an estimate. Your spend limit in Step 1 guarantees it can't run away.

Step 1: Claude API key

This is how Jarvis pays for Claude. It is separate from any claude.ai subscription.

1. Open **platform.claude.com** and sign in (or sign up).
2. Go to **Billing** and add a small amount of prepaid credit (for example $5-10).
3. Go to **Limits** and set a **monthly spend limit** you're comfortable with. This is your safety net.
4. Go to **API Keys > Create Key**, name it **jarvis**, and copy it. It starts with `sk-ant-api`. Keep it in a locked notes app or password manager until Step 3.

Choosing a model (you'll pick it in Step 3)

Model	Best for	Cost
Sonnet 5.5	Everyday assistant use. **Recommended for low cost.**	About half of Opus
Opus 5.5	The smartest answers and trickier multi-step tasks	Highest
Haiku 5.5	The cheapest option; no web search	Lowest

Step 2: Telegram bot (5 minutes, on your phone)

1. Install **Telegram** and create an account.
2. Search for **@BotFather** (it has a blue check mark) and open the chat.
3. Send `/newbot`. Give it a name (e.g. *Jarvis*) and a username that ends in *bot* (e.g. *tony_jarvis_bot*).
4. BotFather replies with a **token** like `123456789:AAF...`. Copy it and keep it with your API key.

You'll get your **chat id** later: the first time you message your bot, Jarvis replies with it.

Path A: Run Jarvis on a cloud server

These steps use Google Cloud from a web browser. A **Windows laptop** is ideal: you install nothing on it, because Google's **SSH** button opens a terminal in a browser window. (From a phone, turn on **Desktop site** in the browser menu.) Google's screens change from time to time, so button names may differ slightly.

A1. Create the account and a budget alert

1. Go to **console.cloud.google.com** and sign up. A card is required for verification. New accounts usually get free trial credit, which covers the first months.
2. Open **Billing > Budgets & alerts > Create budget**. Set it to **$5** so you're emailed if anything costs more than expected.

A2. Create the server

1. Go to **Compute Engine > VM instances** (click **Enable** if asked) and press **Create instance**.
2. **Name:** jarvis. **Region:** us-central1, us-west1 or us-east1 (only these are in the free tier).
3. **Machine type: e2-micro**.
4. **Boot disk > Change:** Ubuntu 24.04 LTS, disk type **Standard persistent disk**, size 30 GB. The default disk type is *not* free, so change it.
5. Leave everything else as is and press **Create**. Wait for the green check mark.

A3. Install Jarvis

Press the **SSH** button next to your VM and click **Authorize**. A terminal window opens. Paste these lines one at a time (Ctrl+V, or right-click > Paste), pressing Enter after each:

```
git clone https://github.com/cmdevsarnia/jarvis.git
cd jarvis
bash deploy/setup-server.sh
```

> **Private repository?**
> If the repository is private, GitHub asks for a username and password. Use your GitHub username, and for the password a **personal access token** (GitHub > Settings > Developer settings > Personal access tokens > Fine-grained, read-only access to this one repository).

A4. Enter your settings

```
.venv/bin/python -m jarvis.configure
```

It asks one question at a time: paste your Claude key, pick model **1** (Sonnet), enter what Jarvis should call you, and paste your Telegram token. Leave the chat id blank for now, and answer **N** to phone calls. Your keys are hidden while you type and stored readable only by you. Then start Jarvis:

```
sudo systemctl restart jarvis
```

A5. Connect Telegram

1. In Telegram, open your bot and send *hello*. It replies with **Your chat id is ...**
2. Back in the SSH window run `.venv/bin/python -m jarvis.configure` again. Press Enter to keep each existing value and paste the chat id when asked.
3. Run `sudo systemctl restart jarvis`.
4. Message the bot: *Jarvis, what's the server's CPU load?* You should get a reply in a few seconds.

Jarvis now starts automatically whenever the server boots. You can close the SSH window. Useful commands: `systemctl status jarvis` (is it running?), `journalctl -u jarvis -n 50` (last 50 log lines).

Free alternative: Oracle Cloud

Oracle's **Always Free** tier includes a VM *and* its public IP at $0, but free accounts have servers **stopped if they stay idle for 7 days**. Jarvis is idle most of the time, so you'd need to upgrade to Pay As You Go (free resources stay free, but you can be billed for anything beyond them). If you try it: create an Ubuntu instance (shape VM.Standard.E2.1.Micro), connect with its **Cloud Shell**, then follow A3-A5 exactly as written.

Path B: Run Jarvis on a Windows PC

1. Install **Python 3.12 or newer** from python.org. On the first installer screen, tick **Add python.exe to PATH**.
2. Get the code: on the GitHub page for the repository, press **Code > Download ZIP** and unzip it to e.g. `Documents\Jarvis`.
3. Double-click **start-jarvis.bat**. The first run installs everything (a few minutes) and then asks your settings, the same questions as Path A.
4. Jarvis opens the HUD in your browser. Use **Microsoft Edge** for the best voice, and allow the microphone.
5. Hold **Space** (or click **MIC**) and talk. Tick **WAKE WORD** to just say *"Jarvis, ..."* hands-free.
6. Connect Telegram as in A5. To change settings later, open a terminal in the folder and run `.venv\Scripts\python -m jarvis.configure`.
7. Optional: edit `config\apps.json` to list apps Jarvis may open for you.

> **Running both?**
> Only one Jarvis can use a Telegram bot at a time. If you move from the cloud server to your PC, stop the server's copy first (`sudo systemctl disable --now jarvis`) or create a second bot.

Jarvis only runs while the window is open and the PC is awake. To start it automatically, add **start-jarvis.bat** to Windows Task Scheduler with the trigger *At log on*.

Add-on: Phone calls

Jarvis can ring your cell and hold a spoken conversation. Do this after Telegram works.

C1. Twilio number

1. Sign up at **twilio.com**. During signup, verify your own cell number.
2. In the Console, **Phone Numbers > Buy a number**: pick one with **Voice** capability (about $1.15/month).
3. From the Console home page, copy your **Account SID** and **Auth Token**.

On a trial account, Twilio can only call your verified number and plays a short trial message first. Upgrading removes that.

C2. Tunnel (so Twilio can reach Jarvis)

1. Sign up free at **ngrok.com**. In the dashboard, copy your **authtoken**, then under **Domains** claim your free static domain (looks like `something.ngrok-free.app`).
2. **Cloud server:** in the SSH window run the line below and paste the token and domain when asked. It keeps the tunnel running permanently.

```
bash deploy/setup-ngrok.sh
```

Windows PC: download ngrok for Windows, then in a second window run the command shown in your ngrok dashboard. It looks like `ngrok http --url=YOUR-DOMAIN 8000`. Keep that window open.

C3. Tell Jarvis and test

1. Run the settings tool again and answer **y** to phone calls. Enter the SID, token, your Twilio number and your cell (both with + and country code, e.g. `+15551234567`), and the public URL `https://YOUR-DOMAIN`.
2. Restart Jarvis.
3. Text your bot: *call me*. Your phone rings. Talk normally. Say *goodbye* to hang up.

Calling *into* Jarvis is off by default, because caller ID can be faked. Jarvis only ever calls your number.

Troubleshooting

Symptom	Fix
Bot never replies	Check `systemctl status jarvis` and the logs. Make sure the token is right and only one copy of Jarvis is running (logs show *409 Conflict* otherwise).
Bot replies with your chat id every time	TELEGRAM_OWNER_CHAT_ID isn't set yet. Run the settings tool and restart.
My API key was rejected	The Claude key is wrong or revoked. Make a new one and run the settings tool.
I'm being rate limited**, or errors about credit**	Add credit or raise the limit at platform.claude.com.
Phone calls are not configured	One of the Twilio settings or PUBLIC_BASE_URL is blank.
Call connects, then *an application error has occurred*	Twilio can't reach Jarvis. Check the tunnel is running and PUBLIC_BASE_URL exactly matches the ngrok domain (https, no trailing slash).
Phone doesn't ring on a trial account	Your cell must be a verified number in Twilio.
HUD says *Access token missing*	Open the exact HUD link printed in the Jarvis window (it includes a token).
Microphone does nothing	Use Chrome or Edge, and allow microphone access when the browser asks.

Safety checklist

- The key you pasted in chat is revoked, and a spend limit is set on platform.claude.com.
- Keys exist only in Jarvis's settings (.env file), which is never uploaded to GitHub.
- TELEGRAM_OWNER_CHAT_ID is set, so Jarvis ignores everyone else.
- Inbound calls stay off unless you understand the caller-ID risk.
- Jarvis can't run arbitrary commands. Each tool is deliberately narrow, so a stolen phone can't take over your PC.
- Calling other people or businesses for you is not built in yet. When it is, it should say it's an AI and only call numbers you approve. AI-voice calls are regulated in the US.

Everyday commands (cloud server)

What	Command
Is Jarvis running?	`systemctl status jarvis`
Recent logs	`journalctl -u jarvis -n 50`
Change settings	`cd ~/jarvis && .venv/bin/python -m jarvis.configure`
Restart	`sudo systemctl restart jarvis`
Update to the latest code	`cd ~/jarvis && git pull && sudo systemctl restart jarvis`
Clear the conversation	Send `/reset` to the bot